

Domain-Specific Pretraining Helps – Until It Does Not: A Case Study on Grassland Weed Recognition

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Abstract. Computer vision models recognising weeds reduce labour and herbicide use in grassland agriculture, but they require large amounts of annotated data. While generic transfer learning is a proven method to mitigate this issue, domain-specific transfer learning has presented itself as a strong alternative. Which of the two to use, and when, remains unclear for grassland weed recognition, as comparisons between these two strategies, across several downstream tasks, remain scarce in the agricultural domain. Therefore, we build the Agro dataset: 305 270 images spanning 96 crop-and-weed species, and use it to pretrain ResNet-50 and ResNet-101 models from scratch. We benchmark our Agro-pretrained backbones against ImageNet-1K V2 pretraining on six target grassland weed species. Our results indicate that in-domain pretraining yields superior representations of the target species. For downstream detection and segmentation, Agro pretraining leads at low data levels (+7.4 pp and +10.1 pp mAP@50 at 20 images for ResNet-50 and ResNet-101), but this advantage decreases then reverses as target data increases.

Keywords: Computer Vision · Domain-Specific Transfer Learning · Agriculture

1 Introduction

Weeds compete with grassland forage grasses and crops for light, water, and nutrients, and their control is one of the most costly tasks in agriculture [22, 24]. Computer vision-based weed detection and segmentation allow for automatic weed management, which lowers manual labour and herbicide needs [22]. The deep-learning-based computer vision models used for these systems, however, require large amounts of training data, and producing the needed annotated agricultural imagery (per-instance bounding boxes and masks) for individual weeds is slow and expensive [24, 55].

A common response to data scarcity is transfer learning; a backbone model is trained on an existing large pretraining dataset and then fine-tuned on the target task. In computer vision that source dataset is often ImageNet [18, 57], since ImageNet-pretrained backbones reliably outperform training from scratch under limited target data availability [29, 65]. The ImageNet dataset, however, contains photographs of everyday objects (cars, dogs, chairs, *etc.*) taken at eye level,

with high object scale variability, which differs greatly from the fine-grained, top-down, and often cluttered imagery typical of agricultural settings. When sufficient target data is available, ImageNet pretraining gains can be limited to faster convergence [25], which leads to the question of whether a pretraining dataset closer to the target domain would transfer better.

Domain-specific pretraining using data gathered from the same domain as the target domain has shown gains in medical imaging [41], and literature reports similar benefits in agriculture [2, 23, 55, 59]. These studies, however, rarely describe how the domain-specific advantage evolves as the number of labelled target images varies over multiple datasets. As a result, which pretraining strategy to choose for grassland weed recognition, and under what conditions, remains poorly documented across both detection and segmentation, multiple backbones, and the full range from low- to high-data availability.

We create the Agro dataset, a classification dataset of 305 270 agricultural images aggregated from open-source crop-and-weed datasets, and use it to pre-train ResNet-50 (RN50) and ResNet-101 (RN101) backbones [27] from scratch. These backbones are then benchmarked against their ImageNet-1K V2 counterparts [62] on six grassland weed species, with no species or genus shared between the Agro dataset and the downstream data.

Contributions. We address the question above: which pretraining strategy to use for grassland weed recognition, and when, across detection and segmentation, multiple backbones, and the full range from low- to high-data availability. We do so by means of two experiments. Specifically: (1) we evaluate representation quality, comparing in-domain and ImageNet embeddings on classes held out during pretraining using standard clustering indices, linear probing, and 2D projections; (2) we benchmark downstream detection and segmentation performance across 15 tasks and both ResNet-50 and ResNet-101 backbones, over a wide range of target data budgets; and (3) we show that the in-domain advantage follows a data-dependent trend established for transfer learning, and that its persistence depends on how closely the pretraining domain matches the target, providing guidance on when each strategy is preferable.

2 Related Work

ImageNet pretraining [18, 57] is a standard pretraining strategy for computer vision. Studies of feature transferability [65] and of the strong correlation between ImageNet accuracy and downstream accuracy [29] have established it as the default across many tasks. He *et al.* [25], however, showed that, with enough target data, models trained from scratch can match ImageNet-pretrained ones. The benefit of pretraining is therefore greatest when target data is limited.

Because generic datasets such as ImageNet differ from specialised domains, pretraining on in-domain data is often the superior strategy. For medical imaging, Raghu *et al.* [49] and Mei *et al.* [41] examined when and why this transfer helps. In agriculture, Al Sahili and Awad [2] showed how pretraining on agricultural

images increased downstream plant disease classification performance, Roggiolani *et al.* [55] showed that pretraining on target-domain data with domain-specific augmentations improved crop and weed segmentation while reducing the labels needed, and Sama *et al.* [59] reported that a plant-specific pretraining dataset reduced overfitting compared to ImageNet. Our study differs in scope: instead of a single dataset or task, we compare supervised in-domain and ImageNet pretraining across two backbones, 15 tasks, and detection and segmentation, and we provide comparisons at numerous labelled target data availability levels.

Deep learning is now the standard approach for weed detection and species recognition [24]. Object detection is performed with models such as Faster R-CNN [54]. Instance segmentation is commonly achieved using Mask R-CNN [26]. Both detection and segmentation tasks are evaluated with COCO-style average precision [31].

The quality of a pretrained model can be evaluated directly from its learned features [47]. A common way of evaluating the quality of those learned features is linear probing [3]: the backbone is frozen and a simple linear classifier is trained on top of the features, providing a direct measure of linear separability. Since these features can be described as high-dimensional data, their geometry can be evaluated for compactness and separability using clustering indices such as the Silhouette score [56], the Calinski-Harabasz index [6], and the Davies-Bouldin index [17]. To be inspected visually, these features can be projected in 2D using t-SNE [36] and UMAP [40]. We use these complementary techniques to compare in-domain and ImageNet features on classes held out during pretraining.

3 Materials and Methods

We compared in-domain pretraining on our Agro dataset against ImageNet-1K V2 pretraining for two model architectures, RN50 and RN101, in two experiments. We first assessed representation quality, then downstream performance on six grassland weed species. This section describes the datasets used, the model architectures (ResNet, Faster R-CNN, Mask R-CNN), and the two experimental setups.

3.1 Data

This section describes the creation of the Agro pretraining dataset and the downstream grassland weed datasets.

Agro Dataset. The Agro dataset contains 305 270 images across 96 species and was created by aggregating open-source crop-and-weed computer-vision datasets [1, 2, 4, 5, 7, 10–12, 16, 19, 28, 30, 35, 37, 43, 46, 48, 50–53, 58, 60, 63]. Datasets in detection or segmentation format were converted to a classification format by cropping the contents of each bounding box, or the square bounding region of each polygon. Datasets already in classification format were added unchanged.

Table 1: The 96 crop and weed species in the Agro pretraining dataset with per-species image counts (305 270 total)

Species	Img.	Species	Img.	Species	Img.
<i>Allium ampeloprasum</i>	2 958	<i>Eleusine indica</i>	425	<i>Parthenium hysterophorus</i>	2 433
<i>Alopecurus myosuroides</i>	2 103	<i>Erigeron bonariensis</i>	30	<i>Persicaria lapathifolia</i>	3 258
<i>Amaranthus palmeri</i>	2 905	<i>Erodium cicutarium</i>	4 043	<i>Persicaria maculosa</i>	2 881
<i>Amaranthus tuberculatus</i>	2 410	<i>Euphorbia geniculata</i>	3 002	<i>Phaseolus vulgaris</i>	4 010
<i>Ambrosia artemisiifolia</i>	1 025	<i>Euphorbia helioscopia</i>	169	<i>Physalis angulata</i>	116
<i>Amsinckia intermedia</i>	989	<i>Euphorbia hirta</i>	571	<i>Plantago lanceolata</i>	3 061
<i>Anagallis arvensis</i>	8 506	<i>Euphorbia hypericifolia</i>	262	<i>Plantago major</i>	4 201
<i>Anchusa arvensis</i>	369	<i>Euphorbia maculata</i>	1 186	<i>Poa annua</i>	5 358
<i>Anoda cristata</i>	61	<i>Euphorbia serpens</i>	2 747	<i>Polygonum aviculare</i>	2 311
<i>Apera spica-venti</i>	2 304	<i>Fallopia convolvulus</i>	1 588	<i>Portulaca oleracea</i>	1 442
<i>Argemone mexicana</i>	854	<i>Fumaria officinalis</i>	139	<i>Raphanus raphanistrum</i>	1 838
<i>Artemisia vulgaris</i>	3 467	<i>Galium aparine</i>	1 548	<i>Rapistrum rugosum</i>	102
<i>Avena fatua</i>	2 317	<i>Geranium molle</i>	3 549	<i>Senna obtusifolia</i>	1 124
<i>Beta vulgaris</i>	14 775	<i>Glebionis segetum</i>	1 128	<i>Sesamum indicum</i>	976
<i>Boerhaavia diffusa</i>	539	<i>Glycine max</i>	1 124	<i>Sida spinosa</i>	613
<i>Brassica napus</i>	3 964	<i>Gossypium hirsutum</i>	2 324	<i>Silene noctiflora</i>	5 430
<i>Bromus sterilis</i>	1 435	<i>Gutierrezia sarothrae</i>	1 016	<i>Sinapis arvensis</i>	3 967
<i>Calotropis procera</i>	5 248	<i>Ipomoea obscura</i>	1 637	<i>Solanum lycopersicum</i>	201
<i>Capsella bursa-pastoris</i>	10 551	<i>Ipomoea purpurea</i>	2 942	<i>Solanum nigrum</i>	5 752
<i>Centaurea cyanus</i>	4 104	<i>Lantana camara</i>	1 064	<i>Sonchus oleraceus</i>	4 096
<i>Chamaecrista pumila</i>	1 306	<i>Lapsana communis</i>	1 298	<i>Sorghum bicolor</i>	1 621
<i>Chenopodium album</i>	6 100	<i>Lepidium didymum</i>	72	<i>Stellaria media</i>	7 698
<i>Chromolaena odorata</i>	1 074	<i>Lolium multiflorum</i>	3 104	<i>Thlaspi arvense</i>	5 791
<i>Clitoria ternatea</i>	678	<i>Lolium perenne</i>	729	<i>Tripleurospermum inodorum</i>	10 603
<i>Commelina benghalensis</i>	1 704	<i>Lolium rigidum</i>	681	<i>Urtica urens</i>	6 790
<i>Convolvulus arvensis</i>	1 256	<i>Malva parviflora</i>	2 100	<i>Vachellia nilotica</i>	1 062
<i>Cryptostegia grandiflora</i>	1 009	<i>Manihot esculenta</i>	5 569	<i>Veronica arvensis</i>	15 856
<i>Cucurbita pepo</i>	1 999	<i>Matricaria chamomilla</i>	5 953	<i>Veronica persica</i>	8 684
<i>Cynodon dactylon</i>	709	<i>Mollugo verticillata</i>	2 326	<i>Vicia hirsuta</i>	3 715
<i>Cyperus rotundus</i>	1 924	<i>Myosotis arvensis</i>	2 735	<i>Viola arvensis</i>	5 743
<i>Digitaria sanguinalis</i>	1 666	<i>Papaver rhoeas</i>	11 910	<i>Zea mays</i>	23 982
<i>Eclipta prostrata</i>	1 119	<i>Parkinsonia aculeata</i>	1 031	<i>Ziziphus mauritiana</i>	1 125

Before inclusion, we removed invalid bounding boxes and polygons and duplicate images, and manually verified the quality of every source dataset (label consistency, picture quality, and foreign objects). The Agro dataset composition is presented in Tab. 1.

Target Grassland Datasets. We evaluated downstream performance on ten publicly available grassland weed datasets and five in-house datasets, together covering six classes. Five of the six classes are distinct species; the sixth pools the *Rumex* genus (*Rumex crispus* and *Rumex obtusifolius*), due to their shared label in our in-house target datasets and the *Rumex WeedEraser* dataset (Tab. 2). We refer to our target classes as “six species” for convenience. To allow for a fair comparison, and to completely exclude the possibility of data leakage, we ensured that no genus or species in the downstream datasets was in the Agro dataset labels. We additionally produced a unified classification version of the downstream grassland datasets (pooling all *Rumex* spp. datasets together), following the same procedure used to create the Agro dataset.

3.2 Models

This section introduces the backbones and the detection and segmentation frameworks used in the study.

Table 2: Downstream grassland weed datasets. *Rumex iNaturalist* [21] and *RumexWeeds* [22] are the only datasets with polygon (mask) annotations and are used for segmentation. The in-house *Bild* and *H** datasets are separate site/date collections.

Species	Dataset	Images	Boxes	Polygons
<i>Rumex</i> spp.	RumexWeeds [22]	5 510	15 519	–
	RumexWeeds_Seg [22]	98	–	340
	Rumex iNaturalist [21]	693	7 166	–
	Rumex iNaturalist_Seg [21]	693	–	7 166
	Rumex WeedEraser [39]	306	765	–
	Rumex (Bild)	575	716	–
	Rumex (HN09)	3 857	5 679	–
	Rumex (HN10)	3 220	4 610	–
	Rumex (HS08)	1 661	2 063	–
	Rumex (HS10)	1 841	2 829	–
<i>Cirsium vulgare</i>	Danish thistles [15]	243	481	–
<i>Cirsium arvense</i>	OPPD [37]	112	1 524	–
<i>Senecio vulgaris</i>	OPPD [37]	146	6 570	–
<i>Senecio madagascariensis</i>	Brownlow Hill Fireweed [14]	20	20	–
<i>Taraxacum officinale</i>	Broadleaf Weeds in Common Couch [13]	78	533	–

ResNet Backbones. All detection and instance-segmentation models were built on the ResNet CNN architecture, a widely used backbone for image feature extraction utilising skip connections [27]. More specifically, we used RN50 and RN101 backbones, which provide two representative model capacities. Each backbone was evaluated under two pretraining strategies: weights obtained entirely from pretraining on the Agro dataset, and standard ImageNet-1K V2 weights.

Faster R-CNN (Detection). Object detection was performed using the Faster R-CNN two-stage detector, which first generates candidate object regions using a region proposal network from a shared convolutional feature map, and then classifies and refines those regions with a second-stage detection head [54]. Each of the four backbone configurations (RN50 / RN101 × Agro / ImageNet-1K V2) was used as the Faster R-CNN feature extractor. The full network was trained end-to-end on each downstream detection dataset.

Mask R-CNN (Instance Segmentation). Instance segmentation was performed using Mask R-CNN, which extends Faster R-CNN with a mask-prediction branch [26]. The same four backbone × pretraining configurations were used. The full network was trained end-to-end as for detection.

3.3 Experimental Setup

We ran two experiments: a representation quality assessment and a downstream performance benchmark. Both are based on the same four backbone configurations, whose pretraining is described below.

Backbone Pretraining. We trained one RN50 and one RN101 from scratch on the full Agro dataset (the six grassland target species are excluded, Sec. 3.1). Alongside their ImageNet-1K V2 counterparts, these are the four configurations (two architectures $\times$ two pretraining sources) used in both experiments.

Representation Quality Assessment. We extracted image embeddings of the grassland species classification dataset (Sec. 3.1) using each backbone. For each of the four models, we measured the separability of the resulting embedding space with three standard clustering metrics: the Silhouette score, the Calinski-Harabasz index, and the Davies-Bouldin index, and visualised the embeddings in 2D with t-SNE and UMAP. Higher Silhouette and Calinski-Harabasz values, and lower Davies-Bouldin values, indicate better separated, more compact clusters. We used 100% of the data to compute the clustering metrics, but capped the 2D projection to 2 000 images per class for readability.

We then applied standard linear-probing to the four backbones. Linear-probing fits a single linear classifier to the frozen features, making it a direct measure of the linear separability of the classes, which is why we use it to assess representation quality instead of downstream performance. We used the six grassland species in classification format as a multi-class classification task, capped at 2 000 samples per class due to extreme imbalance, and as in the downstream performance benchmark, each of the four configurations was run for the same 15 seeds. We report macro-F1 as the primary metric, as it weights all six species equally despite the residual imbalance after capping.

Downstream Performance Benchmark. We evaluated detection and instance-segmentation performance on the grassland datasets provided in bounding-box and polygon/mask format. For each backbone the full network was fine-tuned end-to-end on the target dataset.

To analyse how each pretraining strategy’s performance varies with target data availability, every downstream detection and segmentation dataset was evaluated over a range of training-set sizes rather than only at full size. We defined a single shared grid of data availability levels from the image counts of the downstream datasets (Tab. 2): each dataset’s image count defines one level, and every dataset is evaluated at all levels up to and including its own count. The largest dataset is therefore run at every level, whereas a 243-image dataset is run only at levels ≤ 243 . At a level of N images, N images were used and split across train / validation / test. Every (dataset, level, backbone) condition was repeated for 15 fixed seeds. Because the seed determines the train / validation / test split, the 15 repetitions correspond to 15 independent subsamples and splits. The seed

Table 3: Embedding cluster quality metrics for the four backbone configurations

Metric	Agro RN50	Agro RN101	ImageNet RN50	ImageNet RN101
Macro-Silhouette ($\uparrow$)	0.172	0.179	0.139	0.114
Calinski-Harabasz ($\uparrow$)	1673.0	1947.7	915.0	955.0
Davies-Bouldin ($\downarrow$)	2.118	2.081	2.657	2.938

list was shared across the four backbones, so for a given seed each backbone was trained and evaluated on exactly the same train / validation / test images.

Hyperparameters. All models were trained using the AdamW optimizer [34] ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a weight decay of 1×10^{-4} , a cosine-annealing learning rate schedule decaying to zero, and gradient clipping at a max-norm of 1.0. Mixed precision training (AMP) and ImageNet mean-standard deviation normalisation were applied throughout. For Agro pretraining and linear probing, the input size was 224×224 pixels with aspect-preserving padding, whereas detection and segmentation used a short side of 800 pixels and a long side ≤ 1333 pixels. Agro pretraining employed a learning rate of 5×10^{-5} for 50 epochs with a batch size of 64 and class-weighted cross-entropy loss, augmented by random flips, rotations, and colour jitter. Linear probing used a learning rate of 1×10^{-3} for 5 epochs with a batch size of 64 and standard cross-entropy. Detection and segmentation fine-tuning used a learning rate of 1×10^{-4} for 20 epochs with a batch size of 4, employing the standard multi-task losses of Faster R-CNN and Mask R-CNN, respectively, with no augmentations applied. All experiments were seeded from a single master seed that determined the 15 run seeds, ensuring every run is fully reproducible.

4 Results

4.1 Representation Quality Assessment

Cluster-Quality Metrics. Across all four backbone configurations, Agro pretraining produced better-separated embeddings than ImageNet pretraining on every clustering metric (Tab. 3). Macro-averaged Silhouette rose from 0.114 (ImageNet RN101) and 0.139 (ImageNet RN50) to 0.172 (Agro RN50) and 0.179 (Agro RN101). The Calinski-Harabasz index rose from 915.0 (ImageNet RN50) and 955.0 (ImageNet RN101) to 1673.0 (Agro RN50) and 1947.7 (Agro RN101). The Davies-Bouldin index fell from 2.657 (ImageNet RN50) and 2.938 (ImageNet RN101) to 2.118 (Agro RN50) and 2.081 (Agro RN101). Agro RN101 was the best model across all three metrics.

Per-Class Scores. Since the six downstream datasets are highly imbalanced, we also report per-class Silhouette scores in Tab. 4. Agro pretraining produced

Table 4: Per-class Silhouette for each species for the four backbone configurations (higher is better)

Species	Agro RN50	Agro RN101	ImageNet RN50	ImageNet RN101
<i>Rumex</i> spp.	−0.0529	−0.0211	−0.0527	− 0.0154
<i>Senecio vulgaris</i>	0.1506	0.1597	0.0394	0.0203
<i>Cirsium arvense</i>	0.1629	0.1816	0.1002	0.1012
<i>Cirsium vulgare</i>	0.1967	0.2049	0.2421	0.1489
<i>Senecio madagascariensis</i>	0.2366	0.2248	0.2368	0.2226
<i>Taraxacum officinale</i>	0.3388	0.3223	0.2680	0.2040

Table 5: Linear-probe test performance (percentages, mean $\pm$ SD over 15 seeds)

Metric	Agro RN50	Agro RN101	ImageNet RN50	ImageNet RN101
Macro-F1	98.66 $\pm$ 1.33	97.74 $\pm$ 1.55	91.54 $\pm$ 4.13	93.08 $\pm$ 5.56
Top-1 Accuracy	99.16 $\pm$ 0.23	98.86 $\pm$ 0.31	96.86 $\pm$ 0.46	96.48 $\pm$ 0.51
Macro-precision	99.09 $\pm$ 0.22	97.77 $\pm$ 1.94	96.71 $\pm$ 4.42	94.90 $\pm$ 6.02
Macro-recall	98.38 $\pm$ 2.12	98.04 $\pm$ 2.03	89.15 $\pm$ 4.20	92.16 $\pm$ 5.82

the largest gains on RN101 for *Senecio vulgaris* (0.020 to 0.160), *Taraxacum officinale* (0.204 to 0.322), and *Cirsium arvense* (0.101 to 0.182). For the three remaining species, the highest score across all four backbones was instead achieved under ImageNet pretraining. *Rumex* spp. showed a consistent negative score across all backbones with the best model being ImageNet RN101 (−0.015). On RN50, ImageNet pretraining narrowly outperformed Agro pretraining on *Senecio madagascariensis* (0.2366 to 0.2368) and produced a clear gain on *Cirsium vulgare* (0.1967 to 0.2421).

2D Projections. Figure 1 shows the t-SNE and UMAP projections of the embeddings for the four backbone configurations. For both projections, the Agro-pretrained backbones show visually superior separation for *Senecio vulgaris* and *Cirsium arvense*, cleanly separating the classes into distinct clusters, as opposed to ImageNet-pretrained models. *Rumex* spp. embeddings show as separated sub-clusters across all backbones. The remaining *Cirsium vulgare*, *Senecio madagascariensis*, and *Taraxacum officinale* show no pronounced visual differences across models.

Linear Probing. Table 5 reports linear probing performance for the four backbones. Agro pretraining outperformed ImageNet pretraining on all four metrics: top-1 accuracy (96.86 to 99.16), macro-precision (96.71 to 99.09), macro-recall (89.15 to 98.38), and macro-F1 (91.54 to 98.66); while also being notably more stable, as reflected in its lower SDs throughout.

Table 6 presents the per-class F1 scores for each backbone. Per-class score differences between Agro and ImageNet pretraining stayed approximately within five F1 points for all classes except *Senecio madagascariensis*, where Agro RN50

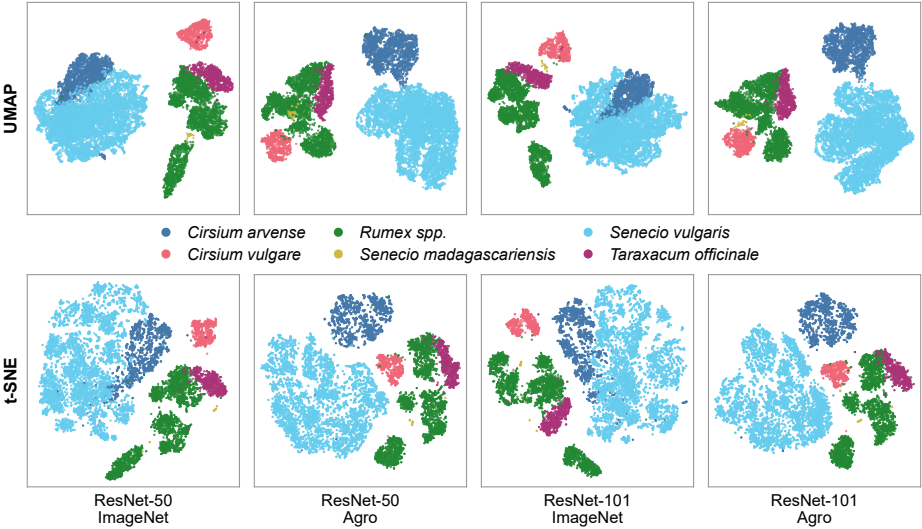


Fig. 1: Per-model 2D t-SNE and UMAP projections of the six grassland species, capped at 2000 images per class for readability

Table 6: Per-class linear probe F1 scores (percentages, mean $\pm$ SD over 15 seeds)

Species	Agro RN50	Agro RN101	ImageNet RN50	ImageNet RN101
<i>Cirsium arvense</i>	99.3 $\pm$ 0.3	99.2 $\pm$ 0.5	94.8 $\pm$ 0.9	94.2 $\pm$ 1.0
<i>Cirsium vulgare</i>	98.8 $\pm$ 0.9	98.4 $\pm$ 1.0	98.3 $\pm$ 1.6	99.0 $\pm$ 0.9
<i>Rumex</i> spp.	99.1 $\pm$ 0.3	98.9 $\pm$ 0.5	98.8 $\pm$ 0.4	98.5 $\pm$ 0.6
<i>Senecio madagascariensis</i>	97.3 $\pm$ 7.0	93.5 $\pm$ 8.4	64.0 $\pm$ 24.4	75.0 $\pm$ 33.5
<i>Senecio vulgaris</i>	99.5 $\pm$ 0.3	99.4 $\pm$ 0.3	96.2 $\pm$ 0.6	95.7 $\pm$ 0.8
<i>Taraxacum officinale</i>	97.9 $\pm$ 1.0	97.1 $\pm$ 1.3	97.1 $\pm$ 2.0	96.0 $\pm$ 2.3

outperformed ImageNet RN50 by 33.3 pp (64.0 to 97.3), and Agro RN101 outperformed ImageNet RN101 by 18.5 pp (75.0 to 93.5). ImageNet pretraining outperformed Agro pretraining on *Cirsium vulgare* only on RN101, by 0.6 pp (98.4 to 99.0); on RN50, Agro led narrowly (98.3 to 98.8).

4.2 Downstream Performance Benchmark

Figure 2 plots the difference in mAP@50 between Agro and ImageNet-1K V2 pretraining (Agro mAP@50 $-$ ImageNet mAP@50), averaged across all seeds and datasets at each training-set size. The RN50 and RN101 lines follow the mean difference; the shaded bands represent the 95% confidence interval of the mean difference; each level is assessed with a paired Wilcoxon signed-rank test. Both backbones follow the same pattern. They begin well above zero (+7.4 pp for RN50, +10.1 pp for RN101) at 20 images, then decrease as target data availability increases.

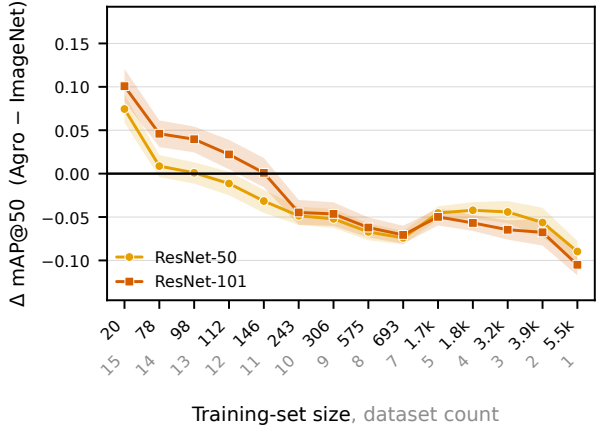


Fig. 2: Agro – ImageNet mAP@50 (fraction scale) difference against training-set size, averaged across datasets

The Agro RN101 kept its advantage over ImageNet for higher data availability levels than the Agro RN50 backbone, crossing into negative performance at 146 images on average, instead of 98. For RN101, the only data availability point at which the Agro – ImageNet difference was nonsignificant is 146, while for RN50, the difference was nonsignificant at 78, 98, and 112 images (being the crossover area). Except for these data points, all levels were highly significant at $p < 10^{-4}$.

The lower (grey) axis row shows the number of datasets reaching each data availability level, from 15 (13 detection, 2 segmentation) to 1. The five non-*Rumex* species reach at most 243 images, so all data points beyond 243 are solely based on *Rumex* spp.

Figure 3 shows the mAP@50 against training-set size per dataset (mean over 15 seeds $\pm$ SD). Red plot borders indicate the two segmentation tasks. Performance increased with data for every dataset. Lines ended at each dataset’s maximum image count (Tab. 2). For non-*Rumex* species, both Agro curves remained above ImageNet curves for all data levels, with the greatest difference at 20 images (deltas between 0.13 and 0.28), and decreasing to 0.05–0.07 at their maximum count. The *Rumex* spp. datasets showed mixed results. Datasets labelled *HN10*, *HS08*, *HS10*, *WeedEraser*, and both *RumexWeeds* (detection) started with a clear Agro pretraining lead that decreased as data availability increased, then reversed for all of them. For *RumexWeeds seg*, the Agro advantage persisted. For the *Bild* dataset, the Agro models outperformed ImageNet models across all data availability levels until all models converged at 575 images. For *HN09* and both *iNaturalist* and *iNaturalist seg* dataset, Agro pretraining underperformed compared to ImageNet across nearly all data availability levels, with only small gains at 20 images.

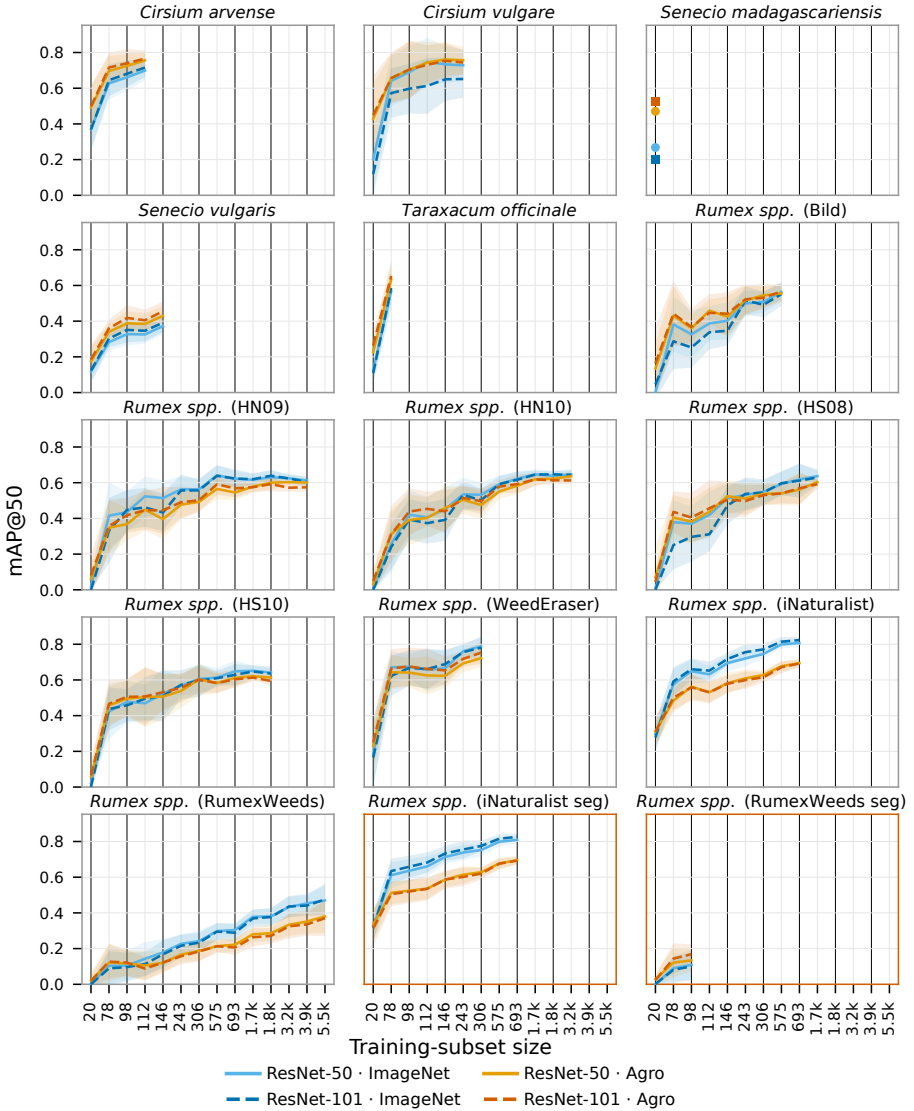


Fig. 3: Per-dataset results grid (fraction scale) comparing the four tested backbones. Plots with red borders represent segmentation tasks (*RumexWeeds Seg*, *Rumex iNaturalist Seg*) and report mask mAP@50. All other plots represent detection tasks and report box mAP@50.

5 Discussion

Representation Quality Assessment. Our representation quality assessment establishes that our Agro pretraining produces representations in which the six held-out target grassland species are more separable than with ImageNet-1K V2.

The clustering quality metrics are consistently superior for Agro-pretrained models: macro-Silhouette, Calinski-Harabasz, and Davies-Bouldin scores (Tab. 3), with Agro RN101 being the best across all models. Per-class Silhouette scores show a slight ImageNet lead for *Rumex* spp. and *Senecio madagascariensis*, but a noticeable ImageNet lead for *Cirsium vulgare* (Tab. 4). The 2D visualisations support these findings; although no quantitative conclusion can be derived, the *Cirsium arvense* and *Senecio vulgaris* clusters are visibly tighter and better separated for all Agro models. The consistently negative Silhouette scores for *Rumex* spp. are likely due to many acquisition sites being aggregated as one class, creating spaced-out sub-clusters under the same label, an effect also visible as separated sub-clusters in the 2D projections (Fig. 1). The linear probe returns a similar pattern: Agro pretraining outperforms ImageNet pretraining on all metrics, and is more stable, as shown by each model’s metric-specific SD (Tab. 5), which matches existing research showing the reduced variance of downstream tasks with improved pretraining [9]. For the per-class linear probe results, the Agro pretraining gains are largest on *Senecio madagascariensis*, which is the class with the fewest samples (20 images). Reduced variance is also observed across all classes, except for *Cirsium vulgare* where ImageNet outperformed Agro pretraining, which also coincides with the largest difference favouring ImageNet in our per-class Silhouette results (0.1967 to 0.2421 on RN50) (Tabs. 4 and 6). These long-tail gains are expected, and in agreement with prior work on rare class recognition, where better models produce the largest gains on classes with the fewest images [33]. Our representation quality assessment has yielded results which are expected for domain-specific transfer learning as it is largely feature reuse for downstream tasks [65], so a backbone pretrained on imagery of top-down weeds and crops already encodes features the downstream tasks rely on. Linear separability under a linear probing protocol is the standard measurement of representation quality [3], and according to that proxy, the Agro features offer higher representation quality.

Downstream Performance Benchmark. Our downstream performance benchmark analyses the adaptability of those features, *i.e.* how that representational advantage behaves as the backbone is fine-tuned on downstream data. On average, we have observed a large advantage of the Agro-pretrained models at low data availability levels. This performance difference decreased as target data increased, and eventually reversed at high data availability levels. Agro pretraining improved perception (detection and segmentation) mAP@50 by +10.1 pp (RN101) and +7.4 pp (RN50) at 20 images; the dataset-pooled advantage falls to zero at 98 target images for RN50 and 146 images for RN101. ImageNet-pretrained models outperform Agro-pretrained models thereafter (Fig. 2). The same low-data gains can be observed for both detection and segmentation tasks, indicating a backbone property rather than being task-head dependent. The crossover behaviour (early Agro advantage to late ImageNet lead) can be observed within datasets (Fig. 3). On RumexWeeds, Agro pretraining leads until 98 images, while that advantage holds up to 306 images on our *Bild* dataset.

No non-*Rumex* datasets display a crossover, which we hypothesise is due to image counts being low compared to *Rumex* spp. datasets. We did, however, observe a decrease in Agro pretraining gains as data availability increased, even for non-*Rumex* datasets. This pattern of an initial performance gain that then decreases and reverses can be explained by our representation quality assessment results and the transfer learning literature. The superior linear separability of Agro features provides better initial performance under label scarcity, already observed by Roggiolani *et al.* [55], who obtained the largest absolute performance difference when using 1% of target data. The decline in transfer-learning benefits as target data grows is well documented [25, 44, 66]; the decreasing-then-reversing pattern for domain-specific pretraining, however, has to our knowledge only been shown in medical imaging [8]. This reversal points to differences in adaptability between Agro and ImageNet pretraining. Training a backbone to separate a set of (pretraining) classes reshapes its deeper layers to compress their input’s numerical rank, retaining only features that are maximally discriminative for the current task; for out-of-distribution applications (note the difference between *domain* and *data distribution* [64]), this loss of representation severely degrades performance [38]. Since our Agro models are pretrained on 96 classes, their higher-level features are much more task-specific and less general than those of ImageNet backbones trained on 1000 diverse classes, explaining the early gains in tasks near their distribution when labels are scarce. Because a fine-tuned model remains in its original loss-landscape basin [45], our specialised Agro models only adapt slowly in directions discarded during pretraining, while generically pretrained ImageNet models are more easily reshaped when data is abundant. An experimental outlier, supporting this interpretation, is the *Rumex* iNaturalist dataset, showing higher ImageNet performance over most data availability levels across detection and segmentation tasks. This dataset was the most dissimilar to our source domain: it contained leaf-instance bounding boxes and masks, while our Agro dataset was built exclusively on whole-plant images, excluding leaf-instance images, further emphasising the need for a near-perfect match if pretraining data is specialised. This is consistent with a narrow source domain needing to match the target domain well, while broader source domains generally transfer better [42].

Interpretation, Limitations, and Future Work. Two aspects of the experimental setup influence how our results should be interpreted. First, we deliberately excluded classes of the same genus/species as the six species in our downstream tasks, leading to a *worst-case* benchmark for our Agro backbones. This choice was made since including these species (we excluded genus-level classes as an additional safety measure), even from completely different sites/datasets, would have likely resulted in greater gain for our Agro backbones but raised data leakage concerns. A real-world application would have selected a pretraining dataset/pretrained model as close as possible to the target domain, including class overlap, which, in theory, should lead to better performance than our conservative lower bound. Second, what limits our specialised backbones at higher

data availability levels is not *domain-specific pretraining* but the *narrowness* of their pretraining label space, as discussed earlier. Increasing the pretraining label space should therefore be the most reliable way of increasing transfer performance. Moreover, adding pretraining scope beyond what might be the target task has minimal side effects on downstream performance [42]. This is confirmed by large-scale pretraining on broadly biological data spanning thousands to hundreds of thousands of classes, which already shows strong results (*e.g.* BioCLIP 2 [20]).

These observations open several directions for future work. The most direct is to build larger domain-specific pretraining libraries and use them at the modest scales seen in practice; today’s state-of-the-art biological foundation models are too computationally expensive for applications that require lightweight models. As the limiting factor across transfer-learning research is a lack of annotated data, contributing to a unified agricultural repository would make such work easier and more accessible. Our findings should also be validated across a wider range of architectures, *e.g.* the ConvNeXt [32] or VGG [61] families, and across broader downstream datasets. These directions follow directly from the limitations of our study. First, we studied only two backbones (ResNet-50 and ResNet-101). Second, at the highest data availability levels only *Rumex* spp. datasets contributed, and at the largest budget only a single dataset did, so the high-data tails are biased toward *Rumex* spp. Finally, all our experiments rest on a single Agro dataset and one pretraining run per architecture, leaving pretraining variance unmeasured.

6 Conclusion

Taken together, our experiments reproduce a pattern already documented for general and domain-specific transfer learning: in-domain pretraining is the better strategy when data availability is low, while generic ImageNet pretraining matches or surpasses it once more data is available. Besides the contributions listed earlier, we have established that the aforementioned pattern also occurs in the agricultural domain. More concretely, Agro pretraining produces more separable and stable embeddings of the six target grassland species, and that advantage translates into a downstream perception gain that is largest under label scarcity, decreases as more data becomes available, and reverses at high data availability. This yields concrete guidance for practitioners. When only tens to low hundreds of downstream images are available, domain-specific pretraining is worth preferring, provided the pretraining domain closely matches the target; a narrow, mismatched source (as with our leaf-instance iNaturalist case) negates that advantage. Generic ImageNet pretraining is the safer choice once downstream data is plentiful. Because the ceiling on domain-specific transfer is the narrowness of the pretraining label space rather than domain specificity itself, the most direct path forward is to broaden domain-specific pretraining datasets so their low-data advantage extends to higher data budgets.

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